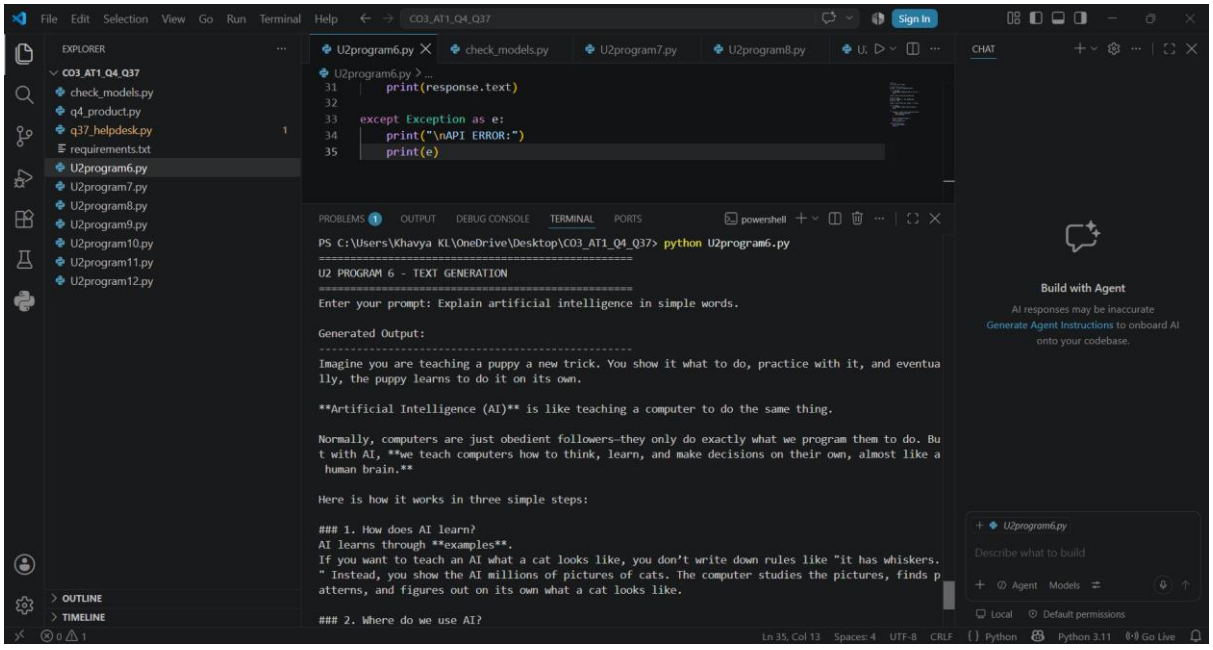
UNIT-2 (6-12 LAB GENAI)

Name: K. L. Khaviya varshini **Reg. No.:** 192472381

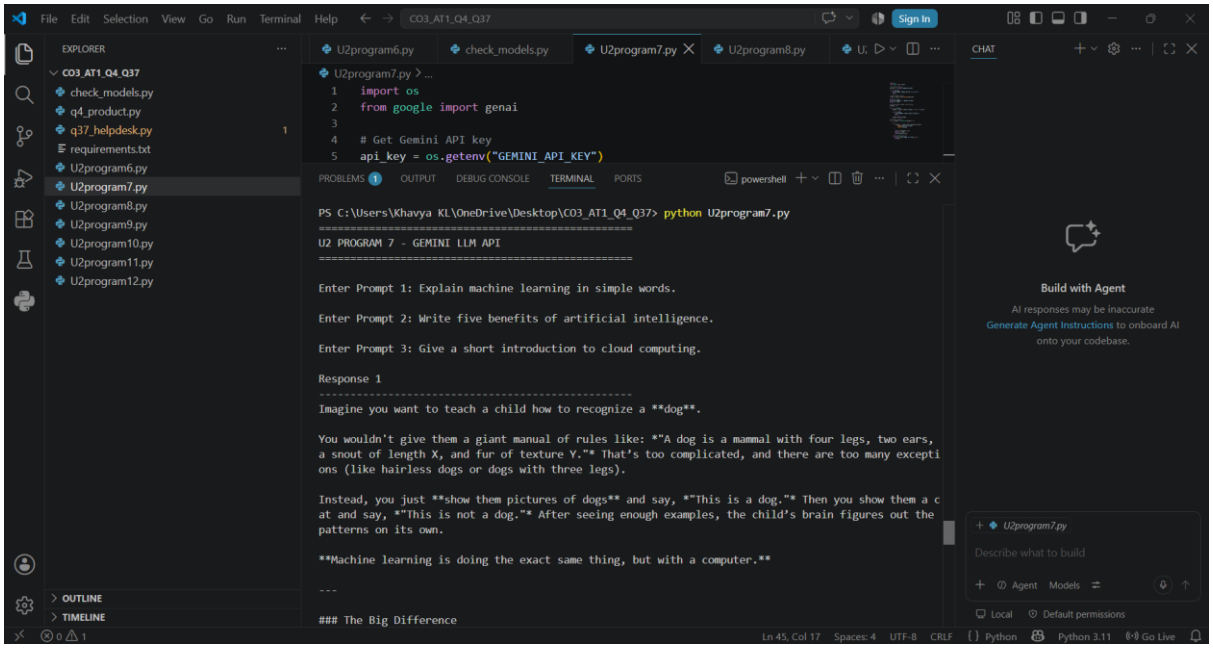
PROGRAM6:

```
import os
from google import genai
client=genai.Client(api_key=os.getenv("GEMINI_API_KEY"))
prompt=input("Enter your prompt: ").strip()
if not prompt:
    print("Error: Prompt cannot be empty.")
else:
    response=client.models.generate_content(model="gemini-3.5-flash",contents=prompt)
    print("\nGenerated Output:")
    print(response.text)
```



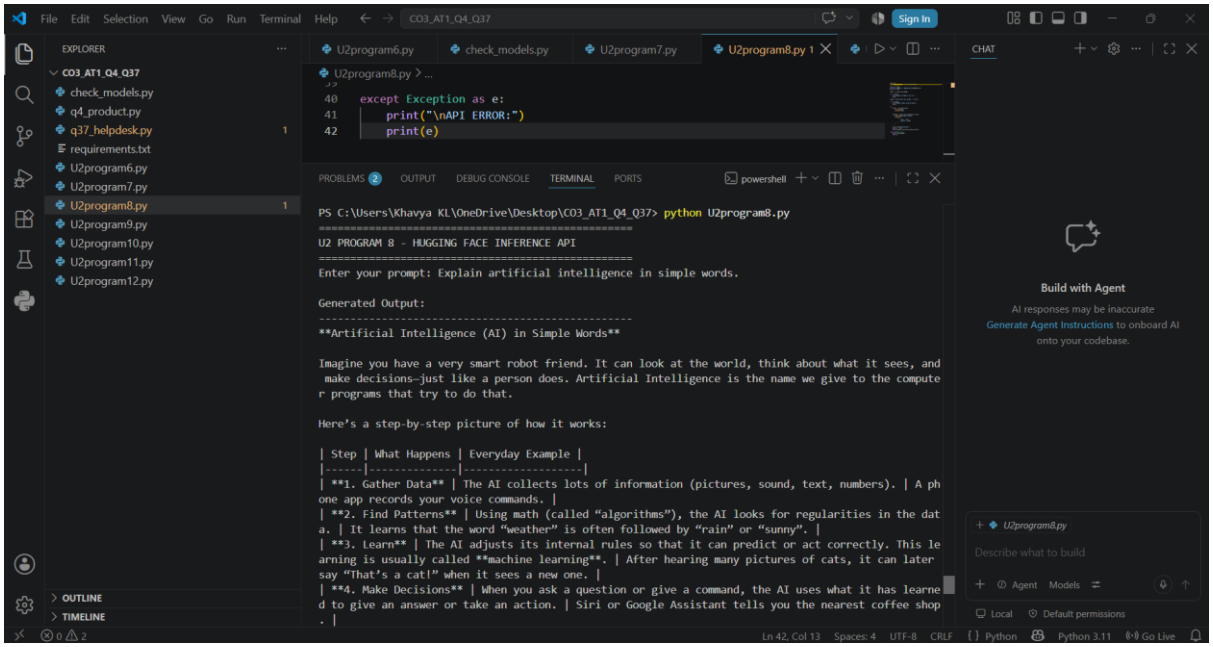
PROGRAM7:

```
import os
from google import genai
client=genai.Client(api_key=os.getenv("GEMINI_API_KEY"))
for i in range(3):
    prompt=input(f"Enter Prompt {i+1}: ").strip()
    if not prompt:
        print("Error: Prompt cannot be empty.")
        break
    response=client.models.generate_content(model="gemini-3.5-flash",contents=prompt)
    print(f"\nResponse {i+1}:")
    print(response.text)
```



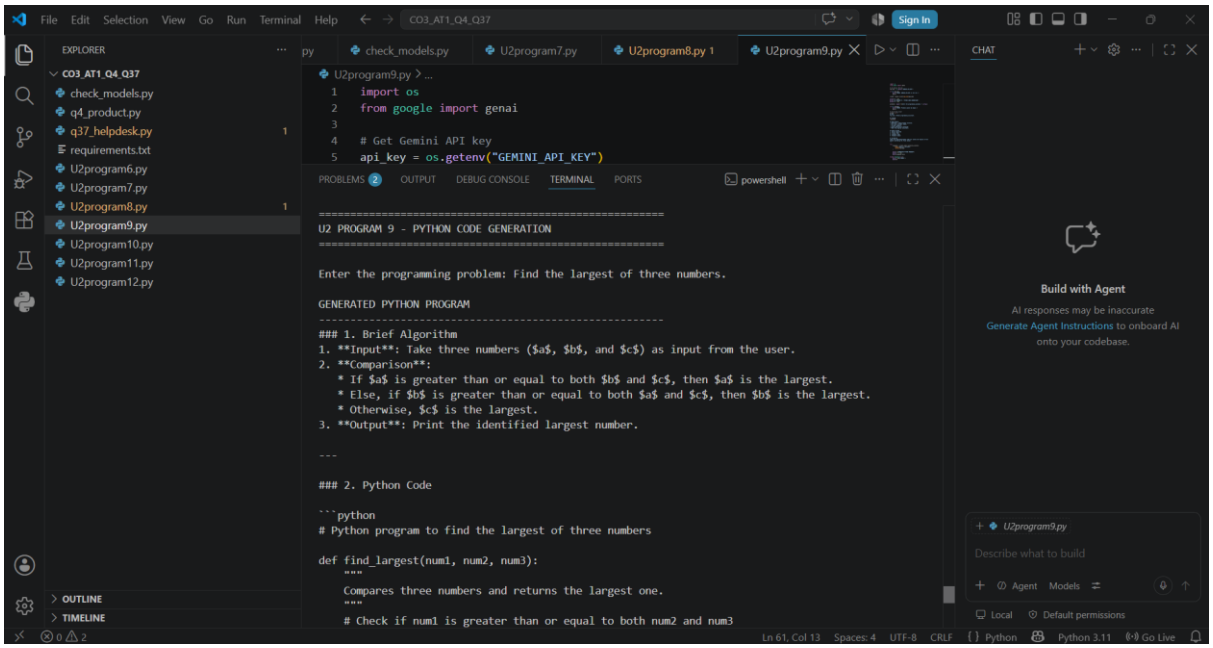
PROGRAM8:

```
import os
from huggingface_hub import InferenceClient
token=os.getenv("HF_TOKEN")
if not token:
    print("Error: HF_TOKEN is not set.")
else:
    prompt=input("Enter your prompt: ").strip()
    if not prompt:
        print("Error: Prompt cannot be empty.")
    else:
        client=InferenceClient(api_key=token,provider="auto")
        response=client.chat.completions.create(model="openai/gpt-oss-120b",messages=[{"role":"user","content":prompt}])
        print("\nGenerated Output:")
        print(response.choices[0].message.content)
```



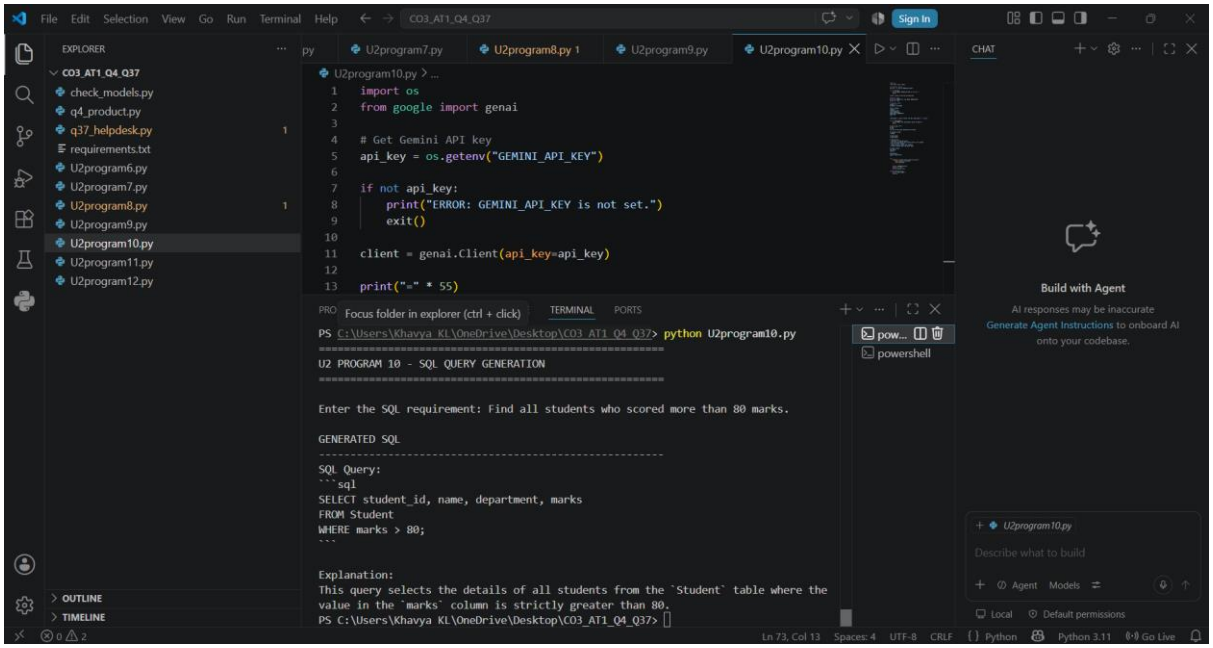
PROGRAM9:

```
import os
from google import genai
client=genai.Client(api_key=os.getenv("GEMINI_API_KEY"))
problem=input("Enter the programming problem: ").strip()
if not problem:
    print("Error: Problem cannot be empty.")
else:
    prompt=f"""Generate a Python program for the following problem.
Problem: {problem}
Requirements:
- Use clear variable names.
- Include comments.
- Make the program executable.
- Provide an algorithm, Python code, example input and example output.
- Check the code for syntax and logical errors."""
    response=client.models.generate_content(model="gemini-3.5-flash",contents=prompt)
    print("\nGenerated Python Program:")
    print(response.text)
```



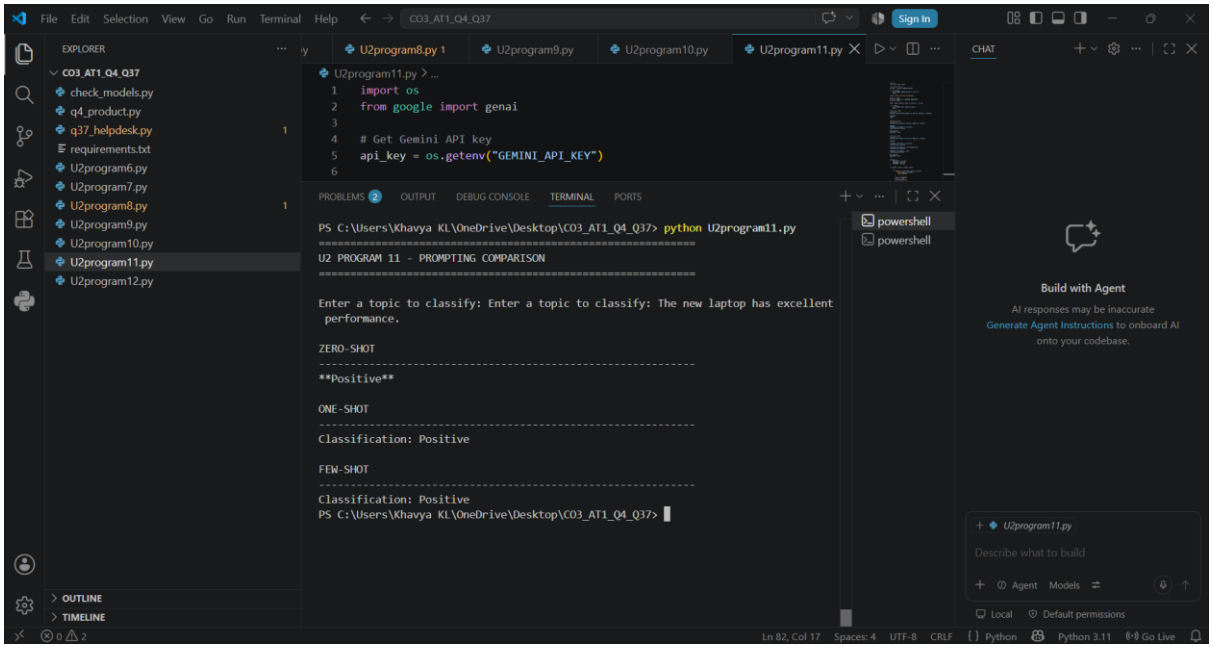
PROGRAM10:

```
import os
from google import genai
client=genai.Client(api_key=os.getenv("GEMINI_API_KEY"))
schema="""Database: CollegeDB
Table: Student
Columns: student_id INT, name VARCHAR(50), department VARCHAR(50), marks INT"""
requirement=input("Enter the SQL requirement: ").strip()
if not requirement:
    print("Error: SQL requirement cannot be empty.")
else:
    prompt=f"""Generate an SQL query using the following database schema.
Schema:
{schema}
Requirement:
{requirement}
Generate a valid SQL query using only the given table and columns and briefly explain it."""
    response=client.models.generate_content(model="gemini-3.5-flash",contents=prompt)
    print("\nGenerated SQL:")
    print(response.text)
```



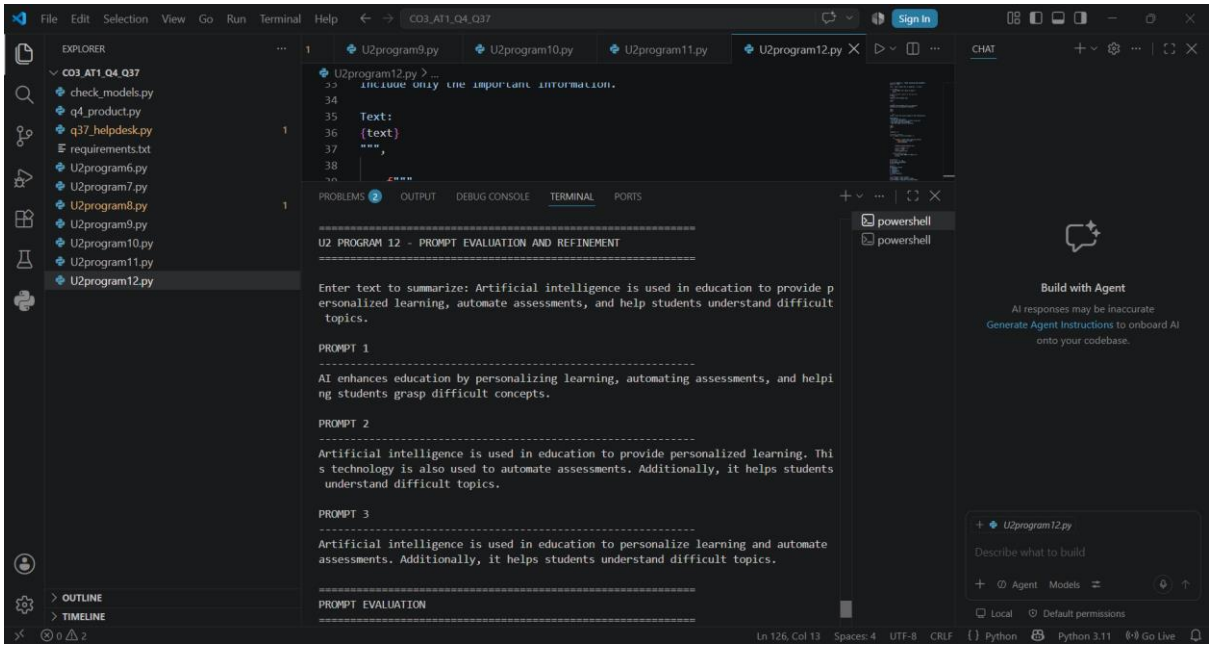
PROGRAM11:

```
import os
from google import genai
client=genai.Client(api_key=os.getenv("GEMINI_API_KEY"))
statement=input("Enter a statement: ").strip()
if not statement:
    print("Error: Statement cannot be empty.")
else:
    prompts=[
        f"Classify the following statement as Positive, Negative, or Neutral:\n{statement}",
        f"Classify the statement as Positive, Negative, or Neutral.\nExample: The product is excellent. Classification: Positive\nStatement: {statement}",
        f"Classify the statement as Positive, Negative, or Neutral.\nExamples:\nThe product is excellent. - Positive\nThe product is disappointing. - Negative\nThe product is okay. - Neutral\nStatement: {statement}"
    ]
    for i,prompt in enumerate(prompts,1):
        response=client.models.generate_content(model="gemini-3.5-flash",contents=prompt)
        print(f"\nPrompt {i}:")
        print(response.text)
```



PROGRAM12:

```
import os
from google import genai
client=genai.Client(api_key=os.getenv("GEMINI_API_KEY"))
text=input("Enter text to summarize: ").strip()
if not text:
    print("Error: Text cannot be empty.")
else:
    prompts=[
        f"Summarize the following text:\n{text}",
        f"Summarize the following text in 3 sentences and include only the important information:\n{text}",
        f"Summarize the following text in exactly 3 sentences. Include the main points, use clear language, avoid repetition, and do not add information not present in the text.\n{text}"
    ]
    for i,prompt in enumerate(prompts,1):
        response=client.models.generate_content(model="gemini-3.5-flash",contents=prompt)
        print(f"\nPrompt {i}:")
        print(response.text)
    refined=f"Create a clear 3-sentence summary. Include the main idea and important points, use simple language, avoid repetition, and do not add outside information.\nText:\n{text}"
    response=client.models.generate_content(model="gemini-3.5-flash",contents=refined)
    print("\nRefined Prompt Output:")
    print(response.text)
```


File Edit Selection View Go Run Terminal Help
CO3_AT1_Q4_Q37
Sign In
EXPLORER
CO3_AT1_Q4_Q37
check_models.py
q4_product.py
q37_helpdesk.py
requirements.txt
U2program6.py
U2program7.py
U2program8.py
U2program9.py
U2program10.py
U2program11.py
U2program12.py
OUTLINE
TIMELINE
U2program9.py
U2program10.py
U2program11.py
U2program12.py
U2program12.py > ...
include only the important information.
Text:
{text}
"""
PROBLEMS 2
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
powershell
powershell
==========
U2 PROGRAM 12 - PROMPT EVALUATION AND REFINEMENT
==========
Enter text to summarize: Artificial intelligence is used in education to provide p
ersonalized learning, automate assessments, and help students understand difficult
topics.
PROMPT 1

AI enhances education by personalizing learning, automating assessments, and helpi
ng students grasp difficult concepts.
PROMPT 2

Artificial intelligence is used in education to provide personalized learning. Thi
s technology is also used to automate assessments. Additionally, it helps students
understand difficult topics.
PROMPT 3

Artificial intelligence is used in education to personalize learning and automate
assessments. Additionally, it helps students understand difficult topics.
==========
PROMPT EVALUATION
==========
CHAT
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI
onto your codebase.
U2program12.py
Describe what to build
Agent
Models
Local
Default permissions
Ln 126, Col 13
Spaces: 4
UTF-8
CRLF
Python
Python 3.11
Go Live